

ZOH Discretization of a Continuous-Time SSM

The continuous-time system, with state $x(t) \in \mathbb{R}^N$ and scalar input $u(t)$:

$$x'(t) = A x(t) + B u(t), \quad y(t) = C x(t). \quad (1)$$

Our data is a discrete sequence, so we need an assumption about the input between samples. The zero-order hold takes it to be piecewise constant, $u(\tau) = u_k$ for $\tau \in [t_k, t_{k+1})$ with $t_{k+1} = t_k + \Delta$. This is the only approximation made.

Multiplying (1) by the integrating factor e^{-At} makes the left side an exact derivative:

$$\frac{d}{dt}(e^{-At}x(t)) = e^{-At}B u(t).$$

Integrating over $[t_k, t_{k+1}]$ and multiplying through by $e^{At_{k+1}}$:

$$x(t_{k+1}) = e^{A\Delta}x(t_k) + \int_{t_k}^{t_{k+1}} e^{A(t_{k+1}-\tau)}B u(\tau) d\tau. \quad (2)$$

This is exact. Now apply the hold: $u(\tau) = u_k$ is constant on the interval, so it leaves the integral, and substituting $s = t_{k+1} - \tau$ gives

$$\left(\int_0^\Delta e^{As} ds \right) B u_k = A^{-1}(e^{A\Delta} - I) B u_k. \quad (3)$$

Writing $x_k := x(t_k)$, the discrete-time system is

$$\boxed{x_k = \bar{A} x_{k-1} + \bar{B} u_k, \quad y_k = C x_k, \quad \bar{A} = \exp(\Delta A), \quad \bar{B} = A^{-1}(\exp(\Delta A) - I) B.} \quad (4)$$

With A diagonal this decouples into N scalar recurrences, $\bar{a}_i = e^{\Delta a_i}$ and $\bar{b}_i = \Delta \varphi(\Delta a_i) b_i$ where $\varphi(z) = (e^z - 1)/z$, so no $N \times N$ matrix is ever formed. Computing φ as `expm1(z)/z` avoids cancellation near $z = 0$; its limit there is 1, which recovers the simplified $\bar{B} = \Delta B$ used in the reference implementation.